

NexaTech Enterprise Network

A small enterprise network designed and implemented in Cisco Packet Tracer, with a focus on network segmentation, controlled inter-VLAN communication, Internet access, and secure device management.

Project Overview

The NexaTech network separates business functions into dedicated VLANs and applies network security controls between departments.

Main objectives

Segment HR, IT, Servers, and Management using VLANs

Provide inter-VLAN routing using Router-on-a-Stick

Automatically assign IP addresses to HR and IT clients using DHCP

Provide static addressing for servers

Control traffic between HR and IT using an extended ACL

Provide simulated Internet access using NAT/PAT

Create a dedicated management VLAN

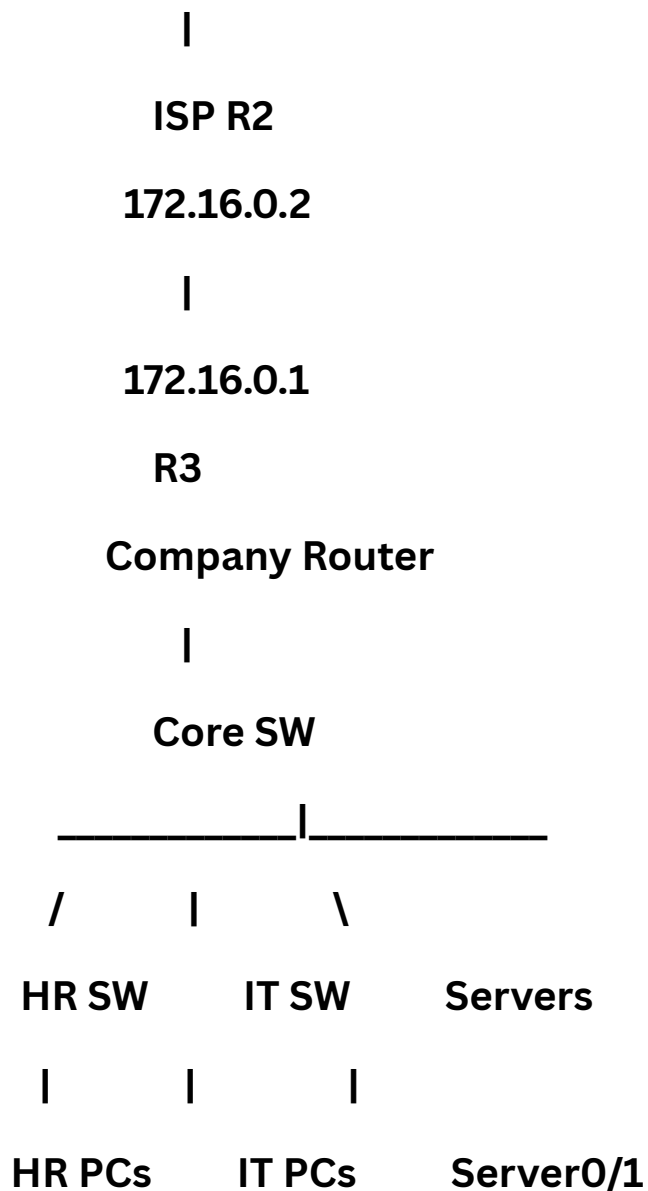
Manage the main router through SSH instead of Telnet

Validate the design through functional and security tests

Topology

Simulated Internet

Cloud-PT



Management VLAN 99

Management PC

VLAN and IP Addressing

VLAN	Purpose	Network	Default Gateway
10	HR	10.10.0.0/26	10.10.0.1
20	IT		

10.10.0.64/26 10.10.0.65 30 Servers 10.10.0.128/26
10.10.0.129 99 Management 10.10.0.192/26 10.10.0.193

Important addresses

Device	Interface / Role	IP Address
R3	VLAN 10 gateway	10.10.0.1
R3	VLAN 20 gateway	10.10.0.65
R3	VLAN 30 gateway	10.10.0.129
R3	VLAN 99 gateway	10.10.0.193
R2	WAN	172.16.0.1/30
R2	Simulated Internet	8.8.8.8
Server1	Static	10.10.0.130
Server0	Static	10.10.0.131
Management PC	VLAN 99	10.10.0.194

Technologies and Features

VLAN Segmentation

Four VLANs were created:

VLAN 10 --- HR

VLAN 20 --- IT

VLAN 30 --- Servers

VLAN 99 --- Management

Trunk links use IEEE 802.1Q and allow the required VLANs between the access switches, core switch, and router.

Inter-VLAN Routing

Router-on-a-Stick is implemented on R3 using subinterfaces:

Gi0/0.10

Gi0/0.20

Gi0/0.30

Gi0/0.99

Each subinterface acts as the default gateway for its VLAN.

DHCP

R3 provides DHCP services for:

HR clients

IT clients

Servers use static addressing.

ACL-Based Access Control

An extended ACL is applied inbound on the HR router subinterface.

Security requirement:

HR must not initiate ICMP communication toward IT

IT must be able to reach HR

HR must still be able to reach permitted networks such as Servers and the simulated Internet

The ACL also permits DHCP client traffic required by HR hosts.

Note: The ICMP-specific rule demonstrates directional ICMP control. It is not a replacement for a stateful firewall for general TCP/UDP session tracking.

NAT/PAT

R3 performs PAT using its WAN interface:

10.10.0.0/24

|

| PAT / overload

v

172.16.0.1

|

v

172.16.0.2

|

Simulated Internet

This allows internal hosts to reach the simulated Internet destination 8.8.8.8.

Secure Management

A dedicated Management VLAN (VLAN 99) is used for network-device management.

R3 is configured for:

Local user authentication

SSH version 2

SSH-only VTY access

Telnet disabled

The Management PC successfully established an SSH session with R3.

Testing and Validation

Test	Expected Result	Result
HR → HR Gateway	Reachable	PASS
IT → IT Gateway	Reachable	PASS
HR → Server	Allowed	PASS
IT → Server	Blocked	PASS
HR → IT	Allowed	PASS
HR → Internet	Allowed through PAT	PASS
IT → Internet	Allowed through PAT	PASS
Management PC → R3	SSH allowed	PASS
Management PC → R3	Telnet blocked	PASS

Testing was performed using ICMP connectivity tests, ACL counters, NAT translation entries, interface status, and SSH/Telnet connection attempts.

Security Controls

The project demonstrates the following controls:

Network segmentation with VLANs

Dedicated management network

Least-privilege communication between departments

Extended ACL traffic filtering

Secure remote administration with SSH

Telnet disabled

Unused access-switch ports disabled where applicable

NAT/PAT at the network edge

Static addressing for servers

DHCP for end-user networks

Repository Contents

NexaTech-Enterprise-Network/

|—— NexaTech-Enterprise-Network.pkt

|—— README.md

|—— configs/

|—— documentation/

|—— screenshots/

The Packet Tracer .pkt file contains the complete simulated network.

How to Use

Install Cisco Packet Tracer.

Download the .pkt file from this repository.

Open it in Packet Tracer.

Inspect the topology and device configurations.

Use the CLI to verify the routing, VLAN, ACL, NAT/PAT, and SSH configuration.

Re-run the connectivity tests listed above.

Project Scope

This is an educational/portfolio network simulation. The Internet connection is simulated inside Packet Tracer using an ISP router and a loopback address.

The configuration demonstrates networking and security concepts and should not be treated as a production

enterprise configuration without additional controls such as centralized AAA, logging/SIEM integration, stateful firewalling, stronger administrative controls, redundancy, and formal change management.

Skills Demonstrated

Cisco IOS configuration

VLANs and trunking

IPv4 subnetting

Router-on-a-Stick

Inter-VLAN routing

DHCP

Extended ACLs

NAT/PAT

SSH

Network troubleshooting

Security testing

Network documentation

Author

Ibrahim Mohamed

Cybersecurity / Networking Student

This project was created as part of my practical development in networking and cybersecurity.